

Building Equivalent Fractions

PhET Fraction Matcher · PhET

Years 7-9

~30 minutes

Mathematics · Number

NOTICE — AI-generated worksheet

This worksheet was drafted automatically from the site's curriculum-mapping data and has not been checked by a human curriculum expert. Please review every task for accuracy, safety and suitability for your specific class before use.

Simulation: <https://phet.colorado.edu/en/simulations/fraction-matcher>

Curriculum links: AC9M7N04 · AC9M7N06 · AC9M8N04

Learning intentions

- I can identify equivalent fractions shown as shapes, numbers and points on a number line.
- I can create my own equivalent fractions using the Lab mode.
- I can use fraction equivalence to solve simple number problems.

Getting started

Open the simulation and start with the Matching Game mode. Choose an early level first (fewer, simpler fractions) before moving to a harder level. Later tasks ask you to switch to Lab mode.

Tasks

1. Play Level 1 of the Matching Game. Pick one matched pair and write both fractions, then explain in one sentence why they are equal even though the numbers are different.

2. Move to a level that includes the number line. Match one pair of fractions using the number line representation. What tells you two fractions are equivalent when you look at a number line?

3. Play several more rounds and record four fraction pairs you match, with the equivalent fraction you matched each one to.

Fraction shown	Equivalent fraction matched to it

4. Using your table, choose one pair and show the multiplication or division you could use to get from one fraction to the other.

5. Switch to Lab mode. Build the fraction $\frac{2}{3}$, then build an equivalent fraction with a larger denominator. Explain how you built it.

6. In Lab mode, try to build a fraction equivalent to $\frac{1}{2}$ that has an odd-numbered denominator. Were you able to do it? Explain why or why not.

7. A recipe uses $\frac{3}{4}$ cup of flour. Without using the simulation, write two other fractions equal to $\frac{3}{4}$ with different denominators, and show your working.

Working:

Extension (optional)

Use Lab mode to build three fractions that are all equivalent to each other but shown using three different representations (e.g. a shape, a number line point, and a numeral pair). Explain what stays the same and what changes between the three.
